Infinite-variation Lévy processes are either strongly eroded or abrupt

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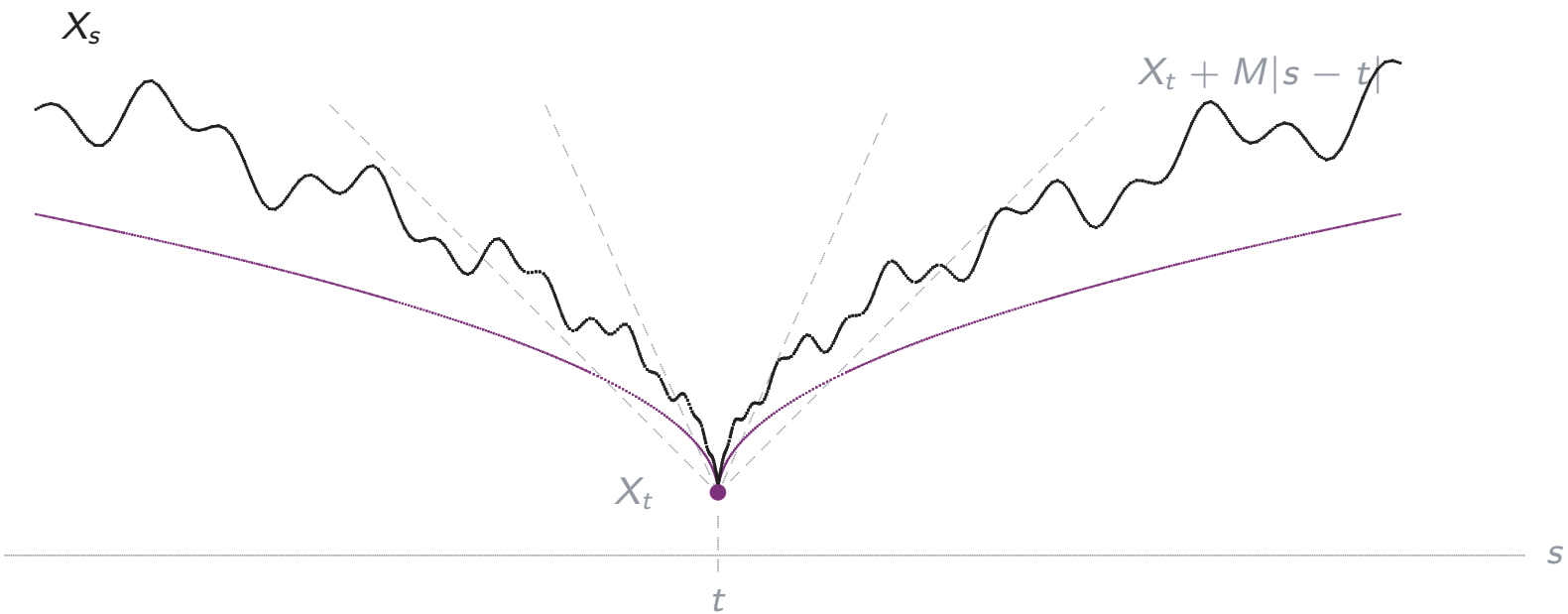
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Abrupt processes

At every local minimum t, almost surely, the one-sided Dini derivatives satisfy

$$\limsup_{h\uparrow 0} \frac{X_{t+h} - X_{t-}}{h} = -\infty, \qquad \liminf_{h\downarrow 0} \frac{X_{t+h} - X_t}{h} = +\infty.$$



Near t, the path rises faster than every fixed linear slope.

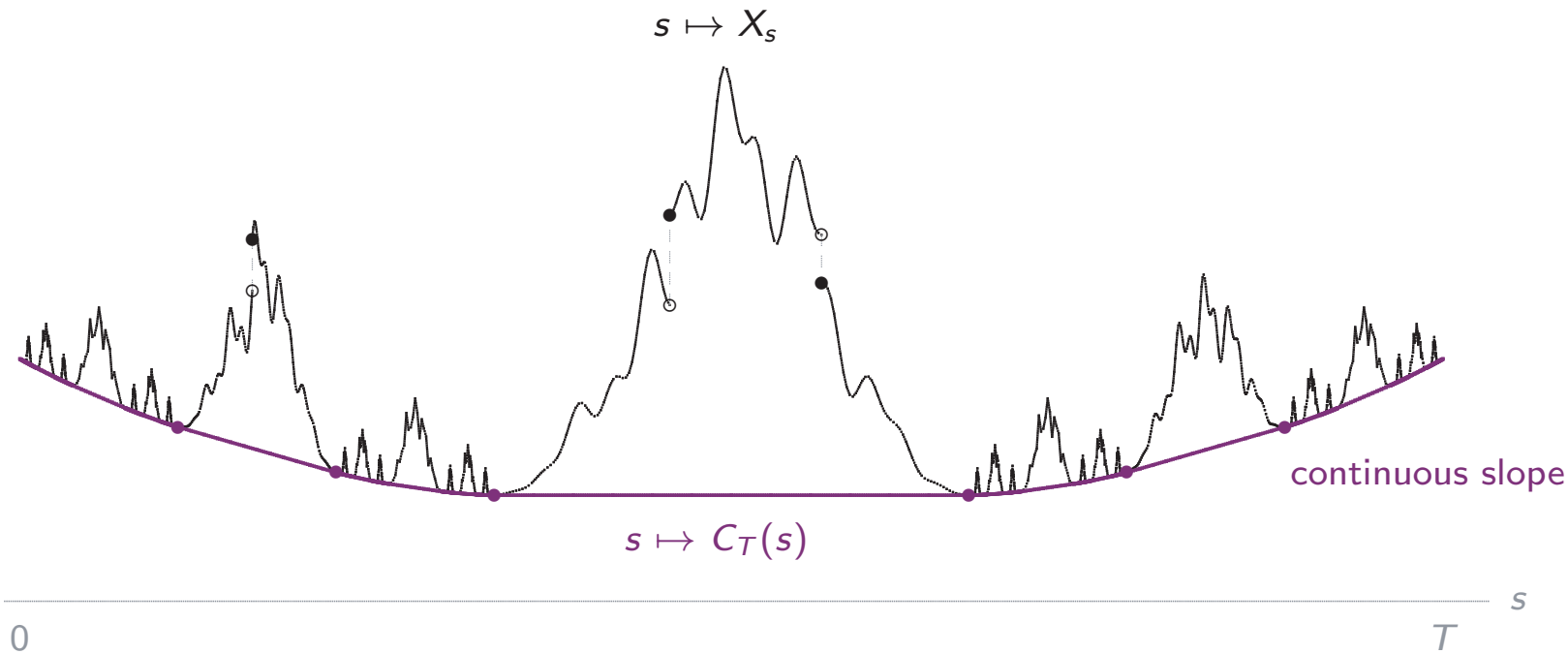
Schematic at a continuous local minimum. The purple curve is a schematic guide.

Vigon, *Stochastic Process. Appl.* 103 (2003), 155–168.

Strongly eroded processes

Let C_T be the greatest convex function below the path on $[0, T]$. The process is strongly eroded if, for every $T > 0$,

$$C_T \in C^1((0, T)) \qquad \text{almost surely.}$$



The convex minorant has affine faces and no corners.

Equivalently, for any linear drift perturbation of X, both Dini derivatives vanish at every local minimum.

González Cázares, Kramer-Bang and Mijatović, *AIHP* 60 (2024), 2352–2381.

The abrupt–strongly eroded dichotomy

Theorem
Every real Lévy process of infinite variation is either abrupt or strongly eroded.

For the convex minorant C_T, exactly one alternative holds:

- Abrupt: piecewise linear, with locally finitely many vertices in $(0, T)$, so $\mathrm{d}C'_T$ is purely atomic with isolated atoms in $(0, T)$.

- Strongly eroded: continuously differentiable on $(0, T)$ and $\mathrm{d}C'_T$ is diffuse singular continuous.

There is no intermediate regime.

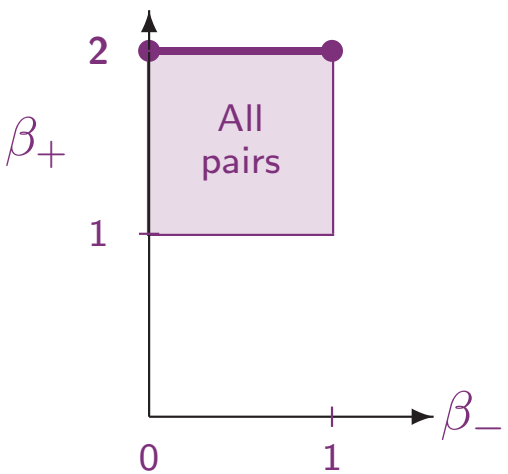
Resolves Conjecture 1.9 in the 2024 paper.

González Cázares, Kramer-Bang and Mijatović, *AIHP* 60 (2024), 2352–2381.

Richness of the strongly eroded class

Theorem
For infinite-variation strongly eroded processes with Lévy measure ν: exactly $(\beta_-, \beta_+) \in [0,1] \times [1,2]$ occur, all with symmetric examples.

Recall (pure jump): $\beta_- > 1 \Rightarrow$ abrupt; $\beta_+ < 1 \Rightarrow$ finite variation.



β_-: Pruitt index

$$\beta_- = \inf\{p > 0 : \liminf_{r \downarrow 0} r^{p-2} \int_{|x|<r} x^2 \nu(\mathrm{d}x) = 0\}.$$

Lower scaling index of small-jump variance.

β_+: Blumenthal–Getoor index

$$\beta_+ = \inf\{p > 0 : \int_{|x|<1} |x|^p \nu(\mathrm{d}x) < \infty\}.$$

Threshold for finite small-jump moments.

Even $\beta_+ = 2$ can be strongly eroded!

- Strictly positive C^∞ Lévy densities on $\mathbb{R} \setminus \{0\}$.
- Bounded C^∞ transition densities, with all derivatives bounded, for $t > 0$.

Gaussian subsequential limits of X_t/t; smooth lacunary bumps.

Orey's example: $(\beta_-, \beta_+) = (0, \alpha)$, $1 < \alpha < 2$, with $\mathfrak{s}_1 \equiv \infty$.

The abrupt–strongly eroded dichotomy

For an interval $I \subset \mathbb{R}$ and real $r \in \mathbb{R}$, define the integrals

$$J_X(I) := \int_0^1 \mathbb{P}(X_t/t \in I)\frac{\mathrm{d}t}{t}, \qquad \mathfrak{s}_1(r) = \frac{1}{2\pi}\int_{\mathbb{R}} \Re \frac{1}{1 + iur - \psi(u)}\mathrm{d}u,$$

where $\psi(u) = \log \mathbb{E}[e^{iuX_1}]$ is the Lévy–Khintchine exponent of X.

Theorem

Let X be a real Lévy process of infinite variation. Exactly one of the following holds:

- Abrupt: $J_X(I) < \infty$ for every bounded interval $I \subset \mathbb{R}$.
- Strongly eroded: $J_X(I) = \infty$ for every proper interval $I \subset \mathbb{R}$.

These conditions can be phrased in terms of $\mathfrak{s}_1$ since

$$J_X(I) < \infty \quad \Longleftrightarrow \quad \int_I \mathfrak{s}_1(r)\mathrm{d}r < \infty.$$

Resolves Conjecture 1.9 of González Cázares, Kramer-Bang and Mijatović, *AIHP* 60 (2024), 2352–2381.

Vigon's conjecture

Since $\mathfrak{s}_1(r) < \infty$ if and only if the Lévy process $X^{(r)} = (X_t - rt)_{t \geq 0}$ hits points (and thus has local time at 0)[1], this connects our conjecture with conjecture of Vigon (2010):

Conjecture
Let X be a real Lévy process of infinite variation. The following conditions are equivalent:

- $\mathrm{s}_1 \in L^1_{\mathrm{loc}}(\mathbb{R})$ — X is abrupt,
- $\mathrm{s}_1(r) < \infty$ for some $r \in \mathbb{R}$ — $X^{(r)}$ hits points for some $r \in \mathbb{R}$,
- $\mathrm{s}_1(r) < \infty$ for all $r \in \mathbb{R}$ — $X^{(r)}$ hits points for all $r \in \mathbb{R}$.

Vigon's conjecture would imply our conjecture; however, it fails.

[1] Kesten, *Memoirs of the AMS* (1969).

Vigon's point-hitting conjecture fails

Theorem
There exists an abrupt symmetric pure-jump Lévy process such that

$$\mathfrak{s}_1(0) = \infty, \qquad \mathfrak{s}_1(r) < \infty \quad \text{for all } r \neq 0, \qquad \mathfrak{s}_1 \in L^1_{\mathrm{loc}}(\mathbb{R}).$$

Pathwise meaning: X itself does not hit points, yet every non-zero linear drift perturbation $X_t^{(r)} = X_t - rt$ does (and admits a local-time field).

Theorem
There exists an abrupt pure-jump process for which the polar set

$$E = \{r \in \mathbb{R} : \mathfrak{s}_1(r) = \infty\}$$

is a compact perfect Cantor set of Lebesgue measure zero.

Constructions use lacunary atomic (or C^∞-smoothed) Lévy measures; exceptional drifts are accumulation points of partial-sum sequences.

The dichotomy: main proof idea

Assume $J_X(I) < \infty$ for one nonempty bounded open interval I.

1. Smoothed characteristics force the running maximum of $|\psi|$ to grow faster than linearly.
2. Local averaging controls exceptional frequencies:

$$\int_{\{u \geq 1:\, |\psi(u)| \leq Mu\}} \frac{\mathrm{d}u}{u} < \infty \qquad (M > 0).$$

3. Elsewhere, resolvent kernels for bounded slope intervals are comparable.

Thus $J_X(J) < \infty$ for every bounded interval J.

Conclusion: Geometry and point hitting

- Geometry: every infinite-variation Lévy process is abrupt or strongly eroded.
- Point hitting: abrupt processes can have a singleton or an uncountable perfect null set of exceptional drifts.

$$X \text{ is abrupt} \iff \mathfrak{s}_1 \in L^1_{\mathrm{loc}}(\mathbb{R}),$$
$$X^{(r)} \text{ hits points} \iff \mathfrak{s}_1(r) < \infty.$$

Open problems

- Characterise strong erosion directly from the Lévy characteristics.
- Classify exceptional drift sets and their Hausdorff dimensions.
- Determine existence and regularity of local times along deterministic curves.
- Express the logarithmic Hausdorff dimension of $\operatorname{supp}(\mathrm{d}C'_T)$ in terms of the Lévy characteristics.